

Case Studies

TM2211 / DA3941 Supply Chain Management

Case 1

Alpha Appliances Ltd. and FlexiTech Electronics are two Sri Lankan manufacturing companies operating in the consumer electronics industry, but they compete using very different operational strategies. Alpha Appliances produces standardized household items such as rice cookers and electric kettles, targeting price-sensitive mass markets. In contrast, FlexiTech Electronics specializes in customizable smart home devices aimed at urban, tech-savvy consumers.

A comparison of key facility-related metrics highlights these strategic differences. Alpha Appliances operates at 92% capacity utilization. FlexiTech, on the other hand, maintains a utilization of 65%. When examining production cost per unit, Alpha reports a lower cost of LKR 2,150, while FlexiTech incurs a higher cost of LKR 3,800. In terms of speed, Alpha has an average flow time of 18 hours per unit, whereas FlexiTech has 6 hours per unit. Flow time efficiency computed for Alpha is 55% compared to FlexiTech's 82%. Additionally, Alpha offers a limited product variety of 25 SKUs, while FlexiTech provides a much wider range of 180 SKUs.

Questions

1. Interpret each of the following metrics for both companies and explain what they reveal about operational priorities:
 - Capacity Utilization
 - Production Cost per Unit
 - Average Flow Time
 - Flow Time Efficiency
 - Product Variety
2. Based on your analysis, identify the supply chain strategy followed by the two companies.
3. Justify your answer using at least three of the metrics given.
4. What trade-offs is each company making in its operational strategy?
5. If market demand becomes highly unpredictable, which company is better positioned to succeed? Explain your reasoning.

Case 2

Nova Retail Solutions is a fast-growing Sri Lankan e-commerce company specializing in fashion and lifestyle products, targeting young, trend-conscious consumers who expect rapid delivery and high product availability. The company follows a responsive supply chain strategy, aiming to quickly adapt to changing customer preferences and demand fluctuations. Nova aims to keep its inventory-related metrics aligned with this approach. They maintain a high fill rate of 96%. Additionally, the fraction of time out of stock is only 3%. The company also holds a relatively high average inventory level to buffer against demand uncertainty and supply delays. Nova's inventory turnover is moderate at 5 times, compared with last year's 9. Similarly, the company reports a C2C (cash-to-cash) cycle time of 75 days. Overall, the company demonstrates a strong commitment to responsiveness, but analysts state that certain inventory inefficiencies remain.

Question

Out of the metrics given, identify which of them align with the responsive supply chain strategy, and which of them don't.

Case 3

QuickMart Lanka is a rapidly expanding online grocery delivery company operating in urban areas such as Colombo and Kandy. The company aims for same-day delivery to customer locations within 6 hours. The management of QuickMart is considering two transportation approaches.

Model A relies heavily on motorbikes and third-party express couriers to ensure fast deliveries. Under this model, the average outbound transportation cost is LKR 420 per order, with an average outbound shipment size of 2.5 units, and an average outbound transportation cost per shipment of LKR 400. Additionally, 75% of deliveries are made using high-speed but expensive modes (bikes and express couriers). The average inbound transportation cost is relatively high at LKR 180 per unit because the company prioritizes frequent supplier replenishment to maintain product availability.

In contrast, Model B emphasizes consolidation by using vans and scheduled deliveries. Here, the average outbound transportation cost is reduced to LKR 250 per order, with a significantly larger average outbound shipment size of 6 units, and a lower average outbound transportation cost per shipment of LKR 220. Only 30% of deliveries use high-speed transport modes, with the majority handled by cost-efficient van routes. The average inbound transportation cost is also lower at LKR 120 per unit, as shipments from suppliers are consolidated into larger batches. However, this model results in longer delivery times, often extending beyond 24 hours.

QuickMart's management must now decide which model better supports its competitive positioning. The challenge is to align transportation decisions with the company's overall supply chain strategy while maintaining profitability.

Questions

1. Analyze the given metrics for models A and B and discuss which model aligns more closely with QuickMart Lanka's strategy.
2. What trade-offs is the company making when choosing between Model A and Model B?

Case 4

Luminex Manufacturing, a Sri Lankan company producing consumer electronics, is evaluating whether to continue producing a key component in-house or outsource it to an external supplier. The component is critical to product performance and demand has been increasing rapidly. The management team is concerned about balancing cost efficiency and responsiveness and has gathered key sourcing-related metrics to support the decision. Their final decision should align with the company's competitive strategy, which increasingly emphasizes fast response to market demand while maintaining acceptable quality standards.

Currently, the in-house production process shows an average purchase (production) cost of LKR 480 per unit, with relatively stable pricing and minimal fluctuation. The average production batch size is 5,000 units, enabling economies of scale. However, the production lead time is 12 days, which limits the company's ability to respond quickly to demand changes. Additionally, the percentage of on-time production is 85%, indicating occasional delays in meeting internal production schedules. The supply quality level is high, with a defect rate of only 1%, ensuring consistent product performance.

Alternatively, an external supplier has offered to provide the component at an average purchase price of LKR 450 per unit, which is lower than in-house production. However, the price shows variability depending on order size, with fluctuations of up to $\pm 8\%$ during peak demand periods. The supplier operates with a smaller average purchase batch size of 2,000 units, allowing more flexibility. The supply lead time is significantly shorter at 6 days, and the percentage of on-time deliveries is 95%, indicating strong responsiveness and reliability in delivery schedules. However, the quality level is slightly lower, with a defect rate of 3%, which may affect final product quality and increase inspection costs.

Questions

1. Based on the metrics, which option out of in-housing and outsourcing is more cost-efficient? Which option is responsive? Explain.
2. What are the key trade-offs between in-house production and outsourcing in this case?
3. Considering Luminex's strategy of improving responsiveness while maintaining acceptable quality, which sourcing option would you recommend?
4. Identify one risk associated with your chosen option and suggest a mitigation strategy.

Case 5

StyleHub Retail is a fast-growing fashion retailer operating both online and through physical stores in Sri Lanka. The company currently follows a high–low pricing strategy, offering high regular prices combined with periodic promotional discounts of up to 30–50%. While this strategy has helped attract customers during sales events, management is concerned about its impact on profitability and supply chain efficiency.

Recent data shows that during promotional periods, the average selling price drops from LKR 4,000 to LKR 2,800, while the average order size increases from 2 units to 5 units per order. Although total sales volume increases significantly, the profit margin declines from 25% to 15% due to heavy discounting and higher incremental variable costs per order (e.g., packaging, delivery, and overtime labor). Additionally, the range of sales per period is highly volatile, with peak demand during promotions being nearly three times higher than normal periods. This has created operational challenges such as stockouts during sales and excess inventory afterward.

Another concern is the Days Sales Outstanding (DSO), which increases from 18 days during regular periods to 30 days during promotions, due to higher use of installment payment options and third-party platforms.

To address these challenges, StyleHub is considering shifting to an Everyday Low Pricing (EDLP) strategy. Under this approach, prices would remain consistently lower throughout the year with minimal promotions. Management has estimated that under EDLP, the average selling price would be LKR 3,200, and demand would become more stable with significantly lower fluctuations in sales volume.

Questions

1. Interpret the following metrics under the high-low pricing method
 - a. Profit margin
 - b. Average selling price
 - c. Average order size
 - d. Incremental variable cost per order
 - e. Day sales outstanding
2. If the company shifts to EDLP, how would the below metrics change?
 - a. Expected average order size
 - b. Expected profit margin
 - c. Expected incremental variable cost per order
3. Identify the key trade-offs between the two strategies

Reflection: How would the decision to shift to EDLP be affected if competitors continue aggressive promotions